Manufacturing Database Creation and Data Migration

Background:
Our manufacturing company produces industrial machinery components. Currently, all records—including raw materials inventory, supplier details, production schedules, machine maintenance logs, quality assurance checks, and customer orders—are managed using a collection of Excel files. As our operations have expanded, this approach has become inefficient, leading to data entry errors, duplicate records, production delays, and significant difficulty in generating accurate reports for decision-making. We now need to migrate all our operational data to a relational database system to improve data integrity, traceability, and overall plant efficiency.

Problem Description:
We require you to design and build a robust, well-structured relational database system that captures all core functionalities of our manufacturing plant. You will need to analyze our current Excel-based data to define the necessary tables and relationships. All existing data must be migrated into this new database, ensuring that data integrity, consistency, and historical accuracy are maintained.

Problems You Need to Solve with Your Database Design:

Lack of Unique Identifiers
Current Issue: We have no guaranteed unique IDs for raw material batches, suppliers, machines, production orders, or individual components. This makes it difficult to trace a specific component back to its origin.

Disconnected Relationships
Current Issue: In Excel, a production order might list a machine and raw materials, but there is no enforceable link to a valid machine in our asset list or an actual batch of material in inventory. This leads to scheduling work on machines that are down for maintenance or using incorrect material grades.

Invalid or Ambiguous Data Entries
Current Issue: Our data suffers from inconsistencies, such as material grades being entered as "SS-304" in one row and "Stainless 304" in another, measurement metrics are Kg for now and KG for some rows.

Unregulated Production Scheduling

Current Issue: Machines are frequently double-booked for different production runs simultaneously. Furthermore, production is sometimes scheduled without verifying if the required raw materials are actually in stock, leading to costly downtime.
Design and implement database check constraints to automatically prevent a machine from being assigned to overlapping time slots and to block the creation of a production order if inventory levels for the required materials are insufficient.

Open Access to Sensitive Production Data
Current Issue: All plant floor staff can currently view all production schedules, customer order details, and material costs, which includes proprietary process information and sensitive financial data.
We need an role-based access control system. For example, machine operators should only see the production tasks assigned to their specific machine. Only line supervisors can view the full production schedule for their line, and only the procurement team can access supplier pricing information.

Disconnected Reporting and Analytics
Current Issue: There is no simple way to generate consolidated reports. Answering questions like "What is our overall equipment effectiveness (OEE) for the month?" or "Which supplier has the lowest defect rate?" requires manually combining data from multiple spreadsheets.
Implement something to generate critical monthly performance reports, such as machine efficiency, raw material consumption rates by production line, and supplier quality scorecards.